

Slide 1: HabotConnect FZCO — Student-LSA Matching Module

Physical Database Architecture, Lineage Locking & Payout Engine

Role: Database Structure Architect

Candidate: [MY FULL NAME] | [MY EMAIL] | [MY PHONE]

Primary Database Engine: PostgreSQL 14+ (ANSI SQL Standard)

Project Focus: 3NF Relational Schema, Declarative Constraints, Immutable Lineage Locks, Fixed-Point Payout Calculations.

Slide 2: Project Context & Business Objectives

Connecting Neurodiverse Students with Certified Learning Support Assistants (LSAs)

- **Mission:** Match children with learning difficulties (ADHD, Dyslexia, ASD) to accredited assistants in UAE.
- **Challenge:** Strict financial auditability, high-frequency mobile telemetry, and complex schedule availability.
- **Objective:** Deliver a deterministic PostgreSQL schema that guarantees zero orphan records and produces the authoritative Monthly Payout Report.

Slide 3: Backward-Engineering Methodology (ED -> LDD -> SD)

Deriving Schema Strictly from Target Business Outputs

- **Target End Document (ED):** 11 required reporting fields for monthly billing and compensation.
- **Logical Design Entities (LDD):** Student Profile, LSA Profile, Session Attendance Log, System Exception Queue.
- **Source Documents (SD):** SD1 (Parent Intake Form), SD2 (HR Onboarding Portal), SD3 (Mobile Telemetry), SD4 (Validation Engine).

Slide 4: Target End Document (ED) Traceability

Authoritative Monthly LSA-Student Match & Payout Report

- 1. Match Allocation ID | 2. Student ID | 3. Student Difficulty Category (STRING_AGG)
- 4. LSA ID | 5. LSA Qualification Level | 6. Compatibility Score (0.0000 - 1.0000)
- 7. Total Session Hours (SUM verified) | 8. Hourly Rate (AED > 0.00) | 9. Total Calculated Payout (Hours * Rate in AED)
- 10. Lineage Trace ID | 11. Immediate Predecessor ID (Lineage Lock FK).

Slide 5: Physical PostgreSQL Schema (11 Normalized Tables)

Comprehensive 3NF Relational Architecture

- **Reference Catalogs:** locations, difficulty_categories, isa_specializations.
- **Core Entities & Junctions:** students, student_difficulties, student_schedules, isas.
- **Matching & Operations:** match_allocations, session_attendance.
- **Lineage & Quarantine:** lineage_audit_ledger, system_exception_events.

Slide 6: Entity Relationship Diagram (ERD) Architecture

Visualizing Physical Relations, Primary Keys, and Foreign Keys

- **8 Business Foreign Keys:** Strict referential integrity on students, difficulties, schedules, LSAs, matches, sessions.
- **10 Lineage Predecessor Locks:** Declarative FKs from all operational tables to lineage_audit_ledger.
- **1 Self-Referencing DAG Link:** Parent-child task execution hierarchy in lineage_audit_ledger.

Slide 7: Golden Rules Enforced at the Database Layer

Physical Constraints Guaranteeing Domain & Financial Integrity

- **Golden Rule 1:** `lsas.hourly_rate` CHECK (`hourly_rate > 0.00`) — physically rejects zero or negative rates.
- **Golden Rule 2:** Universal `trace_id` and `immediate_predecessor_id` across all 11 tables.
- **Golden Rule 3:** `immediate_predecessor_id` NOT NULL REFERENCES `lineage_audit_ledger(execution_id)`.

Slide 8: Immutable Lineage Locking Architecture

Preventing Unanchored Writes and Broken Execution Lineage

- **Concept:** Every row in operational tables must reference a verified task execution node.
- **DAG Enforcement:** Causal execution ordering (Intake -> Onboarding -> Match -> Telemetry -> Report).
- **Tamper Evidence:** Impossible to insert orphan records or unverified mutations into PostgreSQL.

Slide 9: 1NF, 2NF & 3NF Normalization Proof

Decomposing Multi-Valued Attributes and Eliminating Anomalies

- **1NF:** student_difficulties (resolves repeating tags) & student_schedules (resolves repeating slots).
- **2NF:** Composite PK (student_id, difficulty_id) separates difficulty descriptions into difficulty_categories.
- **3NF:** locations & isa_specializations remove transitive dependencies from core profile tables.

Slide 10: Targeted Indexing & Performance Strategy

Optimizing Analytical Queries and Telemetry Lookups

- **Covering Indexes:** `idx_sessions_reporting_covering (match_id, session_start_time) INCLUDE (verified_hours)`.
- **Composite Indexes:** `idx_matches_status_compatibility (match_status, compatibility_score DESC)`.
- **GIN JSONB Indexes:** `idx_exceptions_payload_gin` on raw error packets for sub-millisecond inspection.

Slide 11: Automated Poka-Yoke Mistake-Proofing Tests

10 Negative Test Cases with 100% Database Rejection Rate

- Rate = 0.00 & Negative Rate -> REJECTED (CHECK violation).
- Non-existent Student / LSA / Match FK -> REJECTED (Foreign Key violation).
- Score > 1.0000 or < 0.0000 -> REJECTED (CHECK violation).
- Inverted Session Times & Negative Hours -> REJECTED (CHECK violation).
- Invalid Predecessor ID -> REJECTED (Lineage FK violation).

Slide 12: Monthly Payout Report & Financial Arithmetic

Fixed-Point Precision Currency Calculations in AED

- **SQL View:** view_monthly_isa_student_payout_report with exact rounding $\text{ROUND}(\text{SUM}(\text{hours}) * \text{rate}, 2)$.
- **Numeric Types:** NUMERIC(10,2) for hourly rates and payouts; NUMERIC(5,2) for verified hours.
- **Zero Drift:** Exact fixed-point math prevents IEEE-754 floating point rounding errors.

Slide 13: Polyglot Persistence Strategy

Evaluating Relational vs Time-Series vs NoSQL vs Analytical Warehouse

- **Required Implementation:** PostgreSQL 14+ 3NF Core (ACID consistency, joins, fixed-point math).
- **Recommended Telemetry Tier:** PostgreSQL + TimescaleDB hypertable for 7-day chunking and 90% compression.
- **Recommended Analytical DW:** Google Cloud BigQuery / Snowflake evaluated for multi-terabyte 50M+ row scale.

Slide 14: Realistic Synthetic Demonstration Dataset

Deterministic Multi-Month Operational Scale Validation

- 75 Students Enrolled across 8 UAE geographical zones.
- 25 LSAs Onboarded across 5 specialization domains and 4 competency tiers.
- 60 Active Match Allocations with compatibility scores (0.7000 - 0.9800).
- 225 Operational Sessions across Oct, Nov, Dec 2026; 22 Quarantined Exception Events.

Slide 15: Conclusion & Submission Readiness

Production-Grade Database Structure Architect Deliverable

- **Full Alignment:** Satisfies all HabotConnect database architecture hiring requirements.
- **Clean Structure:** Dedicated SQL, ERD, docs, tests, and presentation files with zero UI clutter.
- **Verified & Defensible:** 100% passing automated test suite and mathematically sound financial reports.
- **Status:** READY FOR HIRING PANEL REVIEW & INTERVIEW DEFENSE.